```
========== Data Analysis & Visualization Program ==========
1. Load dataset
2. Explore data
3. DataFrame operations
4. Handle missing data
5. Generate descriptive statistics
6. Data Visualization
7. Save Visualization
8. Exit
Choose option:  1
Enter your file path (CSV File):  new_insurance_data (2).csv
Data loaded successfully!
```

```
      age     sex     bmi  children smoker  Claim_Amount  past_consultations  \
0     18.0    male  23.210       0.0     no   29087.54313                17.0
1     18.0    male  30.140       0.0     no   39053.67437                 7.0
2     18.0    male  33.330       0.0     no   39023.62759                19.0
3     18.0    male  33.660       0.0     no   28185.39332                11.0
4     18.0    male  34.100       0.0     no   14697.85941                16.0
...    ...     ...     ...       ...    ...           ...                 ...
1333  33.0  female  35.530       0.0    yes   63142.25346                32.0
1334  31.0  female  38.095       1.0    yes   43419.95227                31.0
1335  52.0    male  34.485       3.0    yes   52458.92353                25.0
1336  45.0    male  30.360       0.0    yes   69927.51664                34.0
1337  54.0  female  47.410       0.0    yes   63982.80926                31.0

      num_of_steps  Hospital_expenditure  NUmber_of_past_hospitalizations  \
0         715428.0          4.720921e+06                              0.0
1         699157.0          4.329832e+06                              0.0
2         702341.0          6.884861e+06                              0.0
```

```
3        700250.0       4.274774e+06                          0.0
4        711584.0       3.787294e+06                          0.0
...           ...                ...                          ...
1333    1091267.0       1.703805e+08                          2.0
1334    1107872.0       2.015152e+08                          2.0
1335    1092005.0       2.236450e+08                          2.0
1336    1106821.0       2.528924e+08                          3.0
1337    1100328.0       2.616317e+08                          3.0

      Anual_Salary     region      charges
0     5.578497e+07  southeast   1121.87390
1     1.370089e+07  southeast   1131.50660
2     7.352311e+07  southeast   1135.94070
3     7.581968e+07  southeast   1136.39940
4     2.301232e+07  southeast   1137.01100
...            ...        ...          ...
1333  3.101107e+09  northwest  55135.40209
1334  3.484216e+09  northeast  58571.07448
```

```
1334  3.484216e+09  northeast  58571.07448
1335  3.640807e+09  northwest  60021.39897
1336  4.006359e+09  southeast  62592.87309
1337  4.117197e+09  southeast  63770.42801

[1338 rows x 13 columns]
========== Data Analysis & Visualization Program ==========
1. Load dataset
2. Explore data
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7. Save Visualization
8. Exit
Choose option:  2
========== Explore Data ==========
1. display first 5 rows
```

```
Choose option:  2

========== Explore Data ==========
1. display first 5 rows
2. display last 5 rows
3. display column names
4. display data types
5. display basic info

Enter your choice:  1
    age   sex    bmi  children smoker  Claim_Amount  past_consultations  \
0  18.0  male  23.21       0.0     no   29087.54313                17.0
1  18.0  male  30.14       0.0     no   39053.67437                 7.0
2  18.0  male  33.33       0.0     no   39023.62759                19.0
3  18.0  male  33.66       0.0     no   28185.39332                11.0
4  18.0  male  34.10       0.0     no   14697.85941                16.0

   num_of_steps  Hospital_expenditure  NUmber_of_past_hospitalizations  \
```

```
   num_of_steps  Hospital_expenditure  NUmber_of_past_hospitalizations  \
0      715428.0           4720920.992                              0.0   
1      699157.0           4329831.676                              0.0   
2      702341.0           6884860.774                              0.0   
3      700250.0           4274773.550                              0.0   
4      711584.0           3787293.921                              0.0   

   Anual_Salary     region    charges  
0   55784970.05  southeast  1121.8739  
1   13700885.19  southeast  1131.5066  
2   73523107.27  southeast  1135.9407  
3   75819679.60  southeast  1136.3994  
4   23012320.01  southeast  1137.0110  
```

```
4   23012320.01  southeast  1137.0110
========== Data Analysis & Visualization Program ==========
1. Load dataset
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6. Data Visualization
7. Save Visualization
8. Exit
Choose option:  3

========== Perform DataFrame Operation ===========
1. Filter integer data
2. Filter float data
3. Groupby
4. Minimum Value
5. Maximum Value
```

```
5. Maximum value
Enter your choice:  4
Minimum values:
 age                              1.800000e+01
bmi                              1.596000e+01
children                         0.000000e+00
Claim_Amount                     1.920136e+03
past_consultations               1.000000e+00
num_of_steps                     6.954300e+05
Hospital_expenditure             2.945253e+04
NUmber_of_past_hospitalizations  0.000000e+00
Anual_Salary                     2.747072e+06
charges                          1.121874e+03
dtype: float64
========== Data Analysis & Visualization Program ==========
1. Load dataset
2. Explore data
```

```
========== Data Analysis & Visualization Program ==========
1. Load dataset
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4. Handle missing data
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7. Save Visualization
8. Exit
Choose option:  5
               age          bmi     children  Claim_Amount  \
count  1329.000000  1335.000000  1333.000000   1324.000000
mean     39.310008    30.665112     1.090773  33361.327180
std      14.034818     6.101690     1.201856  15617.288337
min      18.000000    15.960000     0.000000   1920.136268
25%      27.000000    26.302500     0.000000  20768.860390
50%      39.000000    30.400000     1.000000  33700.310675
```
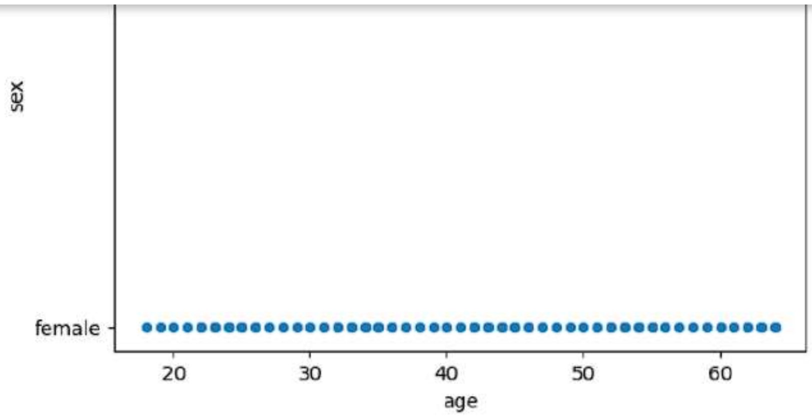

```
75%      51.000000    34.687500     2.000000  45052.331957
max      64.000000    53.130000     5.000000  77277.988480

       past_consultations  num_of_steps  Hospital_expenditure  \
count         1332.000000  1.335000e+03          1.334000e+03
mean            15.216216  9.100047e+05          1.584179e+07
std              7.467723  9.188612e+04          2.669305e+07
min              1.000000  6.954300e+05          2.945253e+04
25%              9.000000  8.471995e+05          4.077633e+06
50%             15.000000  9.143000e+05          7.490337e+06
75%             20.000000  9.716840e+05          1.084082e+07
max             40.000000  1.107872e+06          2.616317e+08

       NUmber_of_past_hospitalizations  Anual_Salary       charges
count                      1336.000000  1.332000e+03  1338.000000
mean                          1.060629  3.696849e+08  13270.422265
std                           0.533583  5.668843e+08  12110.011237
min                           0.000000  2.747072e+06   1121.873900
```

```
2. Explore data
3. DataFrame operations
4. Handle missing data
5. Generate descriptive statistics
6. Data Visualization
7. Save Visualization
8. Exit
Choose option:  6


========== Data Visualization ==========
1. Bar Plot
2. Line Plot
3. Scatter Plot
4. Pie Chart
5. Histogram
6. Stack Plot
```

```
6. Stack Plot
Enter your choice:  3
Enter x-axis column:  age
Enter y-axis column:  sex
```



sex
female
20
30
40
50
60
age

```
========== Data Analysis & Visualization Program ==========
1. Load dataset
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4. Handle missing data
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7. Save Visualization
8. Exit
Choose option:  7
Enter x-axis column:  age
Enter y-axis column:  salary
Enter title:  Age vs Salary
Enter filename (e.g., plot.png):  age_salary.png
Error: 'salary'
========== Data Analysis & Visualization Program ==========
```

```
Choose option:  7
Enter x-axis column:  age
Enter y-axis column:  sex
Enter title:  age vs sex
Enter filename (e.g., plot.png):  age_sex.png

Plot saved as age_sex.png
========== Data Analysis & Visualization Program ==========
1. Load dataset
2. Explore data
3. DataFrame operations
4. Handle missing data
5. Generate descriptive statistics
6. Data Visualization
7. Save Visualization
8. Exit

Choose option:  8

Exiting... Goodbye!
```

```
Enter x-axis column:  age
Enter y-axis column:  sex
Enter title:  age vs sex
Enter filename (e.g., plot.png):  age_sex.png

Plot saved as age_sex.png
========== Data Analysis & Visualization Program ==========
1. Load dataset
2. Explore data
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7. Save Visualization
8. Exit

Choose option:  8

Exiting... Goodbye!
```